

PREDICTION: LEGITIMATE

RISK SCORE: 58 / 100 (MEDIUM RISK)

Scam Probability: 40.10% | Model Confidence: 59.90%

Banking Action: OTP VERIFICATION

Official Recommendation: OTP Verification Required. Moderate risk parameters detected. Challenge customer with SMS OTP or biometric 2FA authentication.

1. Transaction Details & Actual Parameter Values

Transaction ID:	TXN-631446	Transaction Amount:	\$15,000.00
Customer ID:	C123456789	Sender Old Balance:	\$20,000.00
Beneficiary ID:	M987654321	Sender New Balance:	\$5,000.00
Payment Type:	TRANSFER	Receiver Old Balance:	\$500.00
Device Type:	Unknown Proxy	Receiver New Balance:	\$15,500.00
Location Region:	High Risk Region	6h Transaction Velocity:	6 txns
New Beneficiary:	Yes (New)	Prior Fraud History:	No

2. Composite Risk Score Breakdown & Latency Timing

Risk Dimension	Sub-Score	Status & Impact
Model Fraud Probability	40.10%	Weight: 70% (Score: 40.1)
Amount Risk	20.0 pts	Transaction size evaluation
Velocity Risk	20.0 pts	6-hour transaction frequency
Device Risk	25.0 pts	Device profile verification
Location Risk	25.0 pts	Geographic regional risk
Beneficiary Risk	15.0 pts	New beneficiary check
Historical Behaviour	0.0 pts	Prior fraud flag check
Balance Behaviour	0.0 pts	Account wipeout detection
FINAL COMPOSITE RISK SCORE	58 / 100	MEDIUM RISK CATEGORY

Feature Eng: 43.68 ms	Prediction: 1.33 ms	SHAP Engine: 2.95 ms	LIME Engine: 52.31 ms	Rendering: 1152.94 ms	TOTAL: 1253.92 ms
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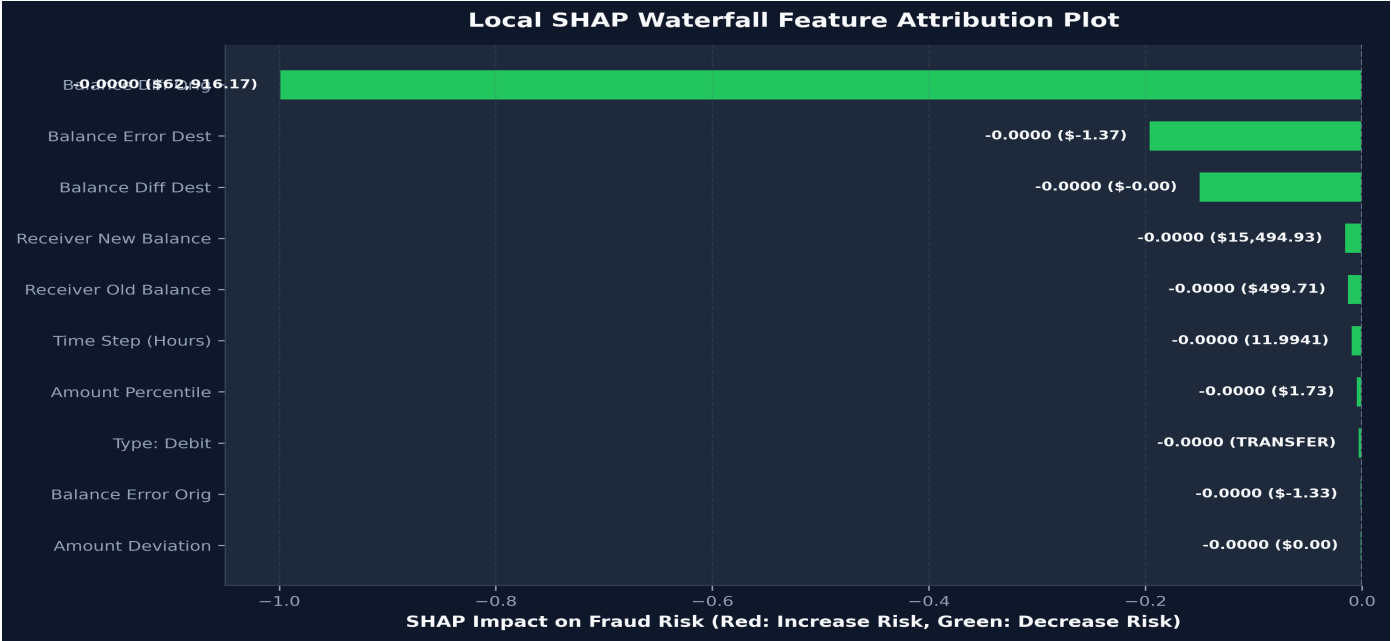
3. Banking-Grade Analyst Narrative Report

Transaction Summary:	Transaction TXN-631446 involving Customer C123456789 transferring \$15,000.00 via TRANSFER to Beneficiary M987654321 on Unknown Proxy from High Risk Region.
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Behaviour Analysis:	Sender account retained \$5,000.00 post-transaction. High velocity anomaly: 6 transactions executed within the last 6 hours. Device anomaly: Payment initiated from unverified device profile ('Unknown Proxy'). Geographic anomaly: Payment originated from high-risk or foreign region ('High Risk Region').
High Risk Indicators:	Elevated Transaction Amount: \$15,000.00 exceeds single transfer threshold.; Rapid Transaction Spike: 6 transfers within a 6-hour window.; Unverified Device Profile: Initiated via Unknown Proxy.; Suspicious Geographic Location: Access from High Risk Region.; Unestablished Beneficiary: No prior transfer history with recipient account.
Safe Indicators:	Account Balance Preserved: Remaining sender balance is \$5,000.00.
Model Explanation:	GuidedGuard Gradient Boosting model evaluated this payment pattern with a scam probability of 40.10% and classified it as 'Legitimate'. The composite banking risk score is categorized under MEDIUM RISK.
SHAP Summary:	Feature contributions align with baseline legitimate profile.
Final Recommendation:	OTP Verification Required: Moderate risk parameters detected. Challenge customer with SMS OTP or biometric 2FA authentication.

4. Top 10 SHAP Feature Contribution Attributions

Feature Name	Actual Value	SHAP Value	Contrib %	Direction	Impact Level
Balance Diff Orig	\$62,916.17	-0.0000	45.47%	Decreases Risk	High
Balance Error Dest	\$-1.37	-0.0000	8.95%	Decreases Risk	Medium
Balance Diff Dest	\$-0.00	-0.0000	6.86%	Decreases Risk	Medium
Receiver New Balance	\$15,494.93	-0.0000	0.75%	Decreases Risk	Low
Receiver Old Balance	\$499.71	-0.0000	0.62%	Decreases Risk	Low
Time Step (Hours)	11.9941	-0.0000	0.48%	Decreases Risk	Low
Amount Percentile	\$1.73	-0.0000	0.25%	Decreases Risk	Low
Type: Debit	TRANSFER	-0.0000	0.19%	Decreases Risk	Low
Balance Error Orig	\$-1.33	-0.0000	0.11%	Decreases Risk	Low
Amount Deviation	\$0.00	-0.0000	0.10%	Decreases Risk	Low



5. Local LIME Decision Rules & Explanations

Decision Rule Condition	Rule Weight	Effect Direction	Domain Explanation
Sender Old Balance = \$20,035.35	+2404.1819	Increases Risk	Feature condition 'Sender Old Balance = \$20,035.35' contributed positively to scam classification.
Receiver New Balance = \$15,494.93	+1859.3316	Increases Risk	Feature condition 'Receiver New Balance = \$15,494.93' contributed positively to scam classification.
Transaction Amount = \$14,970.39	+1796.3870	Increases Risk	Transaction amount is significantly elevated relative to normal threshold.
Sender New Balance = \$5,011.35	+601.3024	Increases Risk	Feature condition 'Sender New Balance = \$5,011.35' contributed positively to scam classification.
Receiver Old Balance = \$499.71	+59.9055	Increases Risk	Feature condition 'Receiver Old Balance = \$499.71' contributed positively to scam classification.
Time Step (Hours) = 11.9941	+1.3793	Increases Risk	Feature condition 'Time Step (Hours) = 11.9941' contributed positively to scam classification.
Type: Payment = TRANSFER	-0.0602	Decreases Risk	Payment type associated with routine account transfers.
Type: Cash Out = TRANSFER	-0.0597	Decreases Risk	Payment type associated with routine account transfers.
Type: Transfer = TRANSFER	+0.0597	Increases Risk	Transfer/Cash Out payment type has higher statistical fraud incidence.
Type: Debit = TRANSFER	-0.0595	Decreases Risk	Payment type associated with routine account transfers.

Local LIME Surrogate Decision Rules Explanation

